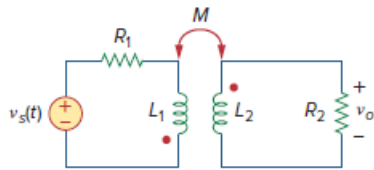


Problem

Using Fig. 16.93, design a problem to help other students better understand how to do circuit analysis with circuits that have mutually coupled elements by working in the s -domain.

Figure 16.93:



Step-by-step solution

Step 1 of 1

Using Fig. 16.62, design a problem to help other students better understand how to do circuit analysis with circuits that have mutually coupled elements by working in the s-domain.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

For the circuit in Fig. 16.62, find $v_o(t)$ for $t > 0$.

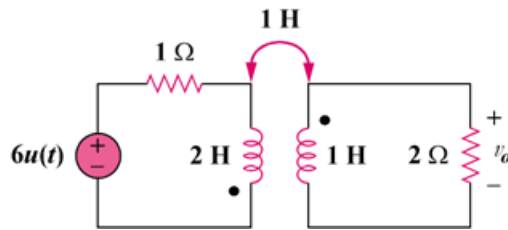
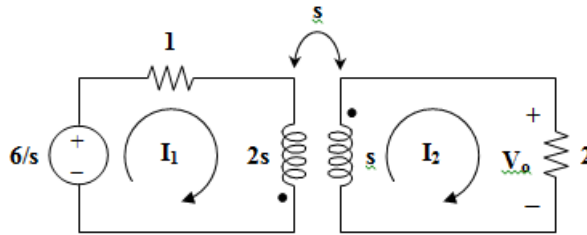


Figure 16.62

Solution

Consider the circuit shown below.



For mesh 1,

$$\frac{6}{s} = (1 + 2s)I_1 + sI_2 \quad (1)$$

For mesh 2,

$$0 = sI_1 + (2 + s)I_2$$

$$I_1 = -\left(1 + \frac{2}{s}\right)I_2 \quad (2)$$

Substituting (2) into (1) gives

$$\frac{6}{s} = -(1 + 2s)\left(1 + \frac{2}{s}\right)I_2 + sI_2 = \frac{-(s^2 + 5s + 2)}{s}I_2$$

$$\text{or } I_2 = \frac{-6}{s^2 + 5s + 2}$$

$$V_o = 2I_2 = \frac{-12}{s^2 + 5s + 2} = \frac{-12}{(s + 0.438)(s + 4.561)}$$

Since the roots of $s^2 + 5s + 2 = 0$ are -0.438 and -4.561,

$$V_o = \frac{A}{s + 0.438} + \frac{B}{s + 4.561}$$

$$A = \frac{-12}{4.123} = -2.91, \quad B = \frac{-12}{-4.123} = 2.91$$

$$V_o(s) = \frac{-2.91}{s + 0.438} + \frac{2.91}{s + 4.561}$$

$$v_o(t) = \underline{2.91[e^{-4.561t} - e^{-0.438t}]}u(t) \text{ V}$$